WHY THE CHECK CAN BE CHEAP

Proof that travels

Checking a reactor-physics number should not cost what producing it did. SMRFORGE seals every result so a reviewer can confirm it was not touched and re-run only the cheap part to see it land where it should. The same seals let an agent carry a whole pipeline forward, one checkable step at a time.

01 · A CHECK THAT COSTS ALMOST NOTHING

You should not re-run the cluster to confirm a number.

A full reproducibility run re-executes the entire pipeline to re-check one result. For a continuous-energy Monte Carlo core that is hours of compute on a cluster, every time anyone wants to be sure. SMRFORGE replaces most of that with two moves a reviewer can make alone.

`smrforge verify <bundle>`

Verify, offline, in milliseconds

Recompute the fingerprints and the signature on your own machine. No engine, no network, no cluster. It confirms the number was not altered since it was sealed, and names the exact section if it was.

`smrforge reproduce <bundle>`

Re-run only the cheap step

A result built from sealed upstream constants re-runs only the downstream deterministic step on those already-verified constants. You check the expensive Monte Carlo once, then never redo it to confirm the numbers that flow from it.

FASTER

Milliseconds, not hours

Verify is a hash check and a signature check. Reproduce re-runs a deterministic solve, not a Monte Carlo campaign.

CHEAPER

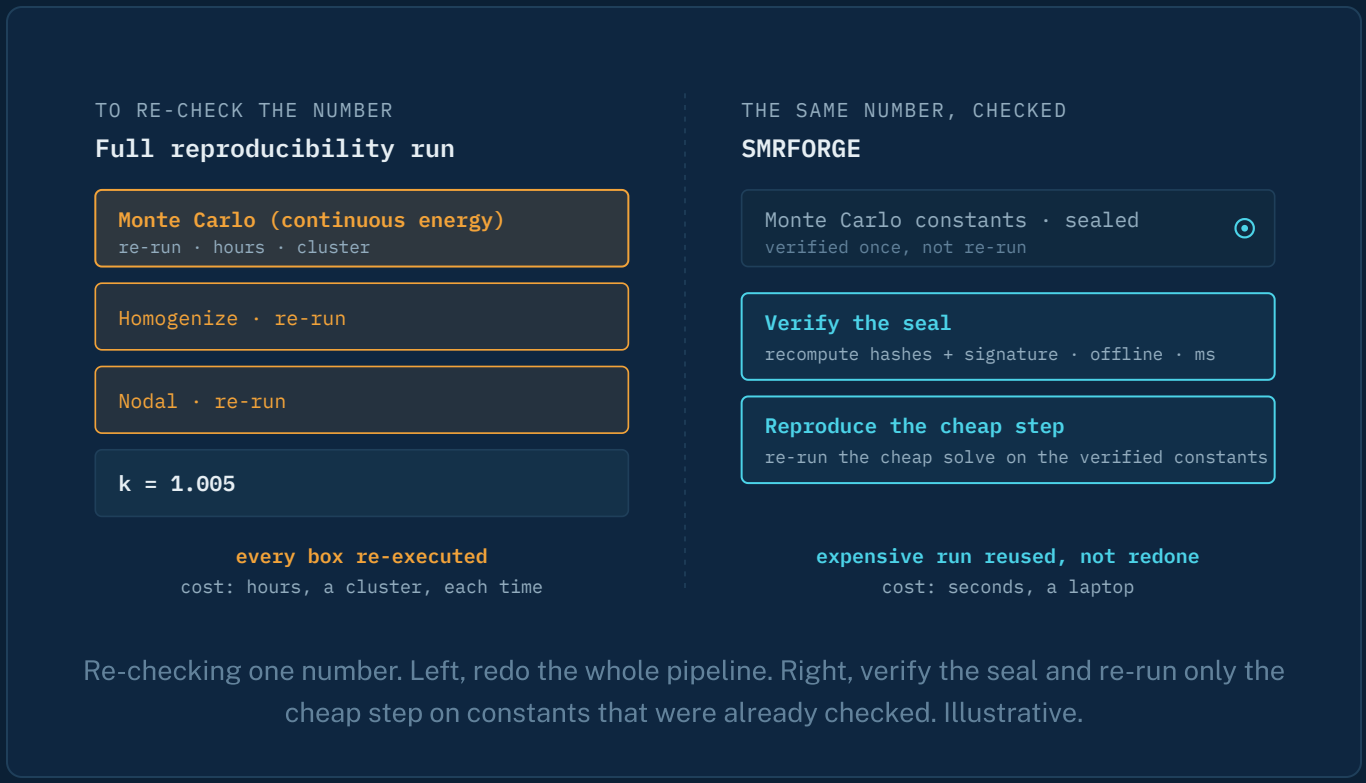
A laptop, not a cluster

The expensive run happens once. Every later confirmation is work a reviewer already has the hardware for.

SMARTER

No trust required

The tolerance is written down before the run. The check stands on its own, offline, without taking the producer's word.



Re-checking one number. Left, redo the whole pipeline. Right, verify the seal and re-run only the cheap step on constants that were already checked. Illustrative.

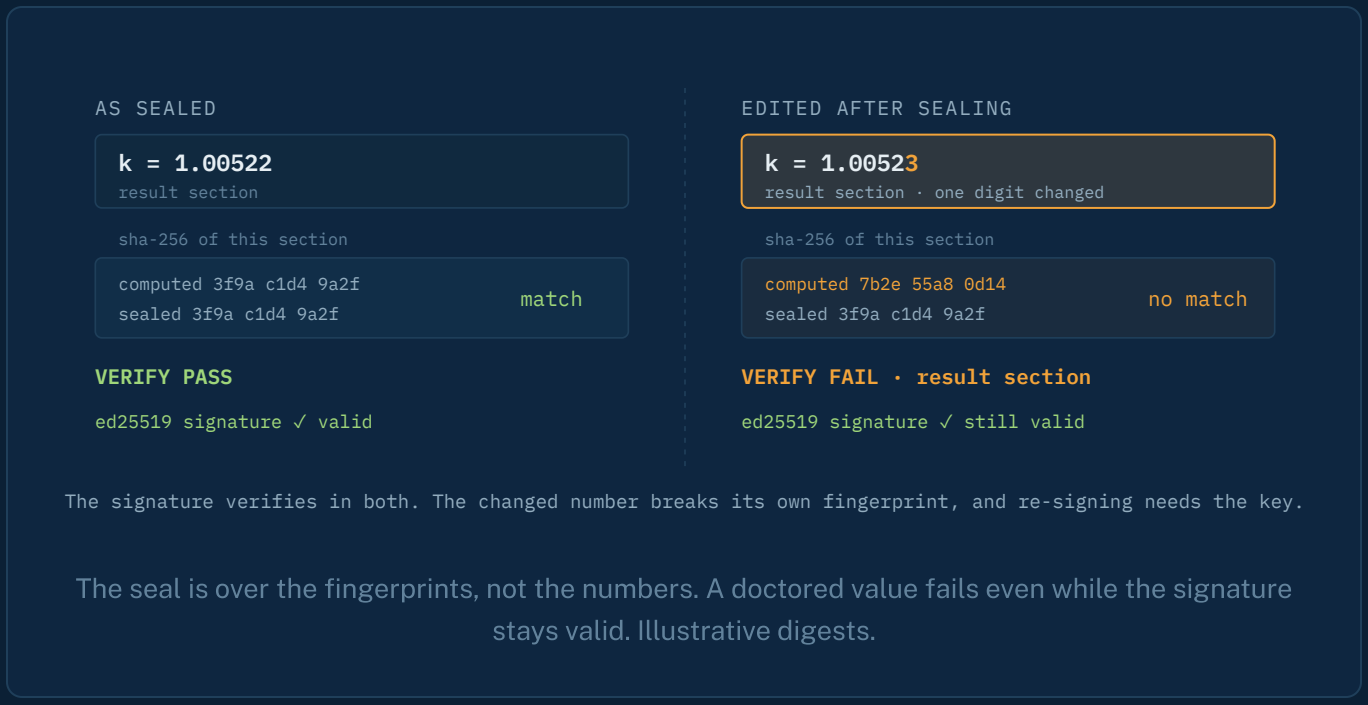
Said plainly. Verify proves custody, not correctness. The original run is still the authority on the physics. SMRFORGE makes the checking cheap, not the computing.

02 · A SIGNED LIE STILL FAILS

The signature can be perfect and the number still gets caught.

The seal covers the fingerprint of each section, and the signature binds those fingerprints. Change one digit of k after a result is sealed and that section stops matching its own fingerprint. The cryptography still checks out, because it signed the

original fingerprint, and the altered number is rejected anyway. To make it pass you would have to re-sign, and you do not hold the key.



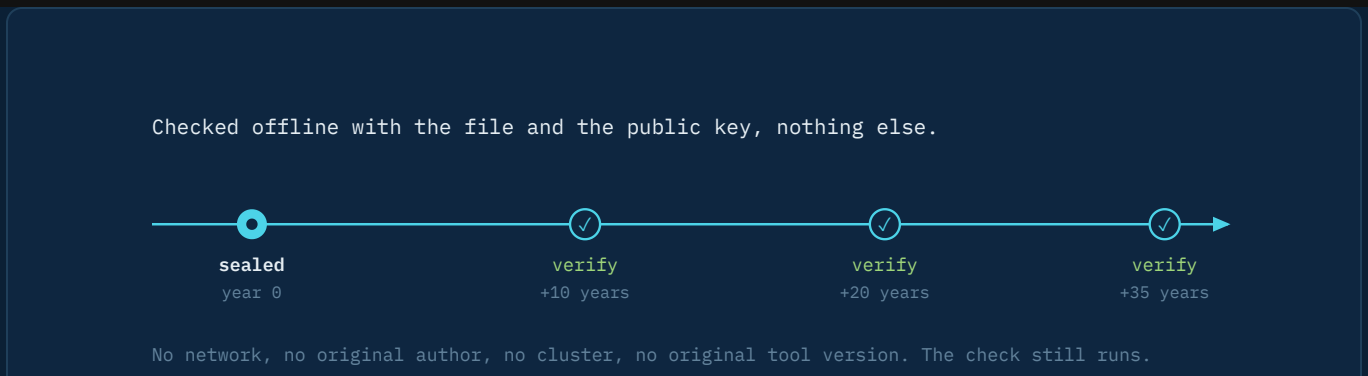
The seal is over the fingerprints, not the numbers. A doctored value fails even while the signature stays valid. Illustrative digests.

You can watch this fail. Open the record, change one digit of k, and run the check. The signature stays green. The number does not. smrforge.io/check.html.

03 · IT STILL CHECKS YEARS FROM NOW

A number that outlives the laptop that made it.

A sealed bundle is a plain file. Checking it needs the file and the public key, nothing else. No network, no original author, no surviving cluster, no copy of the tool that produced it. A reviewer ten years out confirms it the same way a reviewer does today. Reactor projects run for decades and the people change. The number has to stand on its own that whole time, and this one does.



Verify is standalone and offline. The infrastructure that produced the number can be long gone and the record still stands. Illustrative.

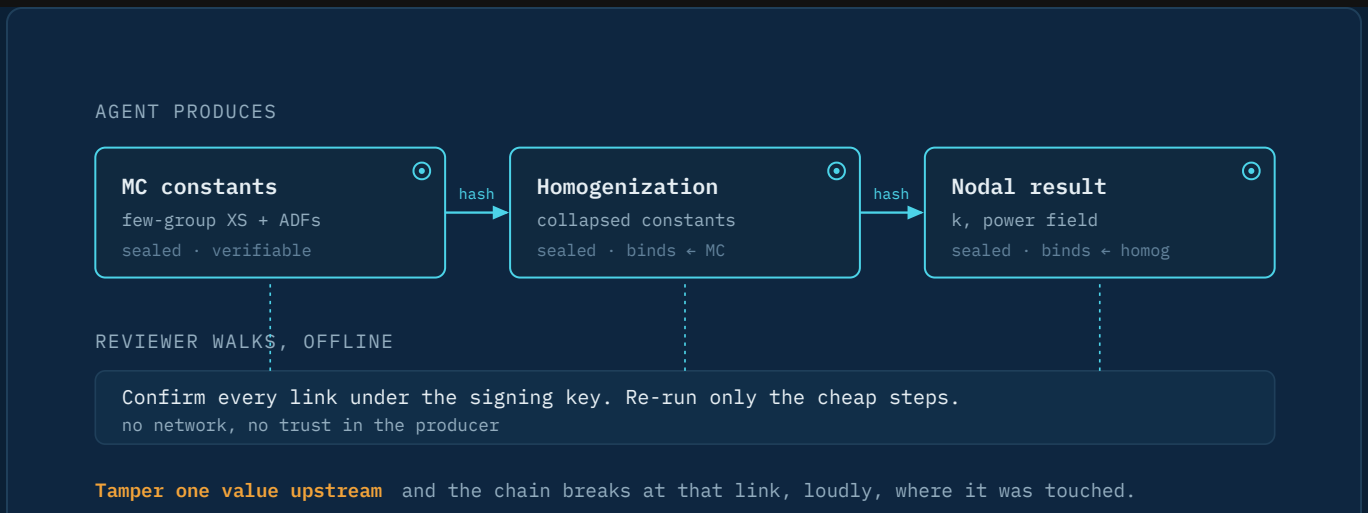
Why it matters here. A licensing record has to be re-checkable for the life of the plant, by people who were not there when it was made. Custody that depends on a live server or the original author is not custody.

04 · A CHAIN BUILT FOR AGENTS

Each step is a file an agent produces and another can check.

A sealed result is just a signed file. A result cites the one it was built from by content hash, so a pipeline becomes a chain of sealed steps, each bound to the last. That shape is exactly what an autonomous pipeline needs.

An agent runs a step, seals it, and binds it to the upstream result. A reviewer, human or agent, walks that chain offline, confirms every link under the signing key, and re-runs only the cheap steps. Change one number upstream and the chain breaks exactly where it was touched.



One result cites the previous by content hash. Agents produce the chain, a reviewer walks it offline. Illustrative.

What is real today. The chaining is live now. A real homogenization-to-nodal chain verifies end to end under a signing key. The autonomous orchestration across a full pipeline is where this is going. AI can produce the number. SMRFORGE is what makes it defensible.

SMRFORGE results are screening-to-reference grade and advisory. They are checked against public criticality benchmarks on open ENDF/B-VIII data, and reproducible by anyone, a lab included. It is not a licensing code, and it says so on every result.

Verify and tamper a real sealed result → smrforge.io/check.html

Break the Seal → smrforge.io/break-the-seal.html

Chris Whalen chris@smrforge.io · smrforge.io